Differences in Circulating Gluconeogenic Substrates During Short-Term Fasting in Men, Women, and Children

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To determine whether the differences in fasting glucose responses of men, women and children could be related to differences in glucoregulatory hormone secretion or availability of circulating gluconeogenic substrates, 20 adults (10 men and 10 women) were fasted for 86 hr and 15 children (6.1 ± 0.8 yr, mean ± SE) were fasted for 30 hr. Circulating concentrations of glucose, ketone bodies, potential gluconeogenic substrates and glucoregulatory hormones were determined at frequent intervals. In the postprandial state (1–14 hr fasting), substrate and hormone concentrations were similar in all groups with the exception of plasma glutamine concentrations which were higher in men (640 ± 20 μM) than in women or children (490 ± 20 μM and 480 ± 50 μM, respectively, $p < 0.01$ for both). Following 30 hr fasting children had the lowest glucose (53 ± 3 mg/dl) and alanine (167 ± 17 μM) concentrations and men had the highest (72 ± 3 mg/dl and 279 ± 16 μM, respectively) whereas those of women were intermediate (64 ± 3 mg/dl and 226 ± 19 μM, respectively). Following 30 hr of fasting, betahydroxybutyrate concentrations were highest in children and lowest in men (the children, 3.7 ± 0.4 mM; women, 1.7 ± 0.2 mM and men 0.9 ± 0.2 mM, $p < 0.02$ between all groups). An inverse relationship ($p < 0.001$) was observed between beta-hydroxybutyrate and glucose concentrations throughout the fast for each group ($r > 0.92$). Although plasma cortisol concentrations were higher in women and children when compared to those of the men, no differences in growth hormone, glucagon, or insulin concentrations were observed among the three groups studied. Differences in plasma substrate responses to fasting among the groups studied may reflect differences in glucose requirements among men, women and children, as well as the relative availability of both gluconeogenic substrate and ketone bodies.

YOUNG CHILDREN develop relative or absolute hypoglycemia following 30–42 hr of fasting[1–3] when compared to adults[4,5] who rarely achieve these mean levels of glycemia, even following several days (or in the case of obese subjects, for several weeks) of caloric deprivation. Differences in circulating concentrations of hormones and metabolic fuels have been reported between fasting in men and women. Women are reported to have lower plasma glucose concentrations but higher plasma free fatty acid and glucagon concentrations than men during short-term fasting.[4,5,6]

One might expect similar or higher plasma glucose concentrations in fasting women than men since the increased plasma glucagon should increase hepatic glucose production and increased availability of FFA should decrease glucose utilization. Therefore, the observed differences in the circulating concentrations of these compounds, FFA and glucagon, do not explain the lower plasma glucose concentrations observed in women. Decreased availability of potential gluconeogenic substrate has been suggested to contribute to the absolute or relative hypoglycemia in children[2,7] and adults.[8,9] Whether substrate availability or alteration in secretion of glucoregulating hormone could contribute to the difference in plasma glucose concentrations in men, women and children has not been systematically investigated. Therefore, the following studies were undertaken to determine whether some of the metabolic differences in response to short-term fasting in men, women and children could be related to differences in hormone secretion, or availability of gluconeogenic or ketogenic substrate, as reflected in their circulating plasma concentration.

MATERIALS AND METHODS

Patient Population

Ten normal healthy adult male and ten female subjects and 15 prepubertal children were admitted to the Clinical Research Unit (Table 1). None of the subjects had a family history of diabetes; all had fasting glucose concentrations < 110 mg/dl and were consuming a weight-maintaining diet prior to study. The adults were consuming at least 200 gm carbohydrate per day for three days prior to the study. None of the subjects had a history compatible with hypoglycemic episodes or documented hypoglycemia. The 15 children were selected from patients who had had symptoms which were compatible with hypoglycemia and were fasted as part of their clinical evaluation. None previously had documented hypoglycemia, nor developed symptoms or signs of hypoglycemia during their evaluation.

Study Protocol

All individuals were fasted following an evening meal (1700–1800 hr). Bloods were obtained between 1800–2000 hr, 0800 hr, and subsequently at 4 hr intervals. The adults were fasted for 86 hr and the children for 30 hr. The children were monitored throughout the fast as previously described.[2] Four to five milliliter of blood were

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Table 1. Demographic Data of the Subjects Studied

	Age (yr)	Height (cm)	Weight (kg)
Men			
(n = 10)	24.0 ± 1.9*	181 ± 2	73.7 ± 2.7
Women			
(n = 10)	23.7 ± 1.1	168.0 ± 1.3	58.3 ± 1.7
Children			
(boys/girls = 6/9)	6.1 ± 0.8	114.2 ± 5.9	21.5 ± 2.5

*All values represent mean ± SEM.

drawn into a syringe containing Na heparin (0.1 ml) at each sampling and divided into three tubes; (1) 2 ml of blood were placed in 750 IU of Trasylol (FMA Pharmaceuticals), (2) 0.5 ml was precipitated immediately with an equal volume of cold 3M perchloric acid, and (3) the remaining blood placed in an additional tube. All samples were placed immediately on ice, centrifuged at 4°C, and the supernatant stored at −80°C until assayed.

Analytical and Statistical Methods

Plasma glucose, alanine, glutamine and glutamate and blood lactate, pyruvate, beta-hydroxybutyrate and acetoacetate were determined microfluorometrically as previously described.[10,11] Amino acid determinations were performed on Spinco-Beckman 119 Amino Acid Analyzer using standard physiologic methodology.[12] Plasma amino acid concentrations were determined at all sampling times in six men and six women and on the 0, 14, 22, and 30 hr fasted samples of five children. Plasma glucagon,[13] insulin,[14] and growth hormone[15] concentrations were determined by radioimmunoassays and cortisol by a protein binding assay[16] on the plasma samples collected with Trasylol.

The fasting profile data were evaluated statistically by a multivariant analysis of variance (AOV). Differences between groups were analyzed by the appropriate multiple range test utilizing an IBM computer. The statistical differences between data at specific time points within each group were determined by a paired two-tailed Student's t test, and difference between groups at a specific time by a non-paired Student's t test. All data are expressed as mean ± SE.

RESULTS

Glucose

Glucose concentrations were similar in the three groups studied during the first 18 hr of fasting (Fig. 1). By 30 hr fasting, the plasma glucose concentration was lowest in children and highest in the men and the groups were statistically different one from another (children, 53 ± 3 mg/dl and women 64± 3 mg/dl, $p < 0.05$; children versus men, 72 ± 3 mg/dl, $p < 0.001$; and men versus women, $p < 0.001$). Utilizing analysis of variance, the glucose concentrations were higher in men than in women ($p < 0.001$) and higher in the adults than in the children ($p < 0.05$) throughout the entire period of study.

Beta-hydroxybutyrate and Acetoacetate

Postprandial beta-hydroxybutyrate and acetoacetate concentrations were similar and rose progressively

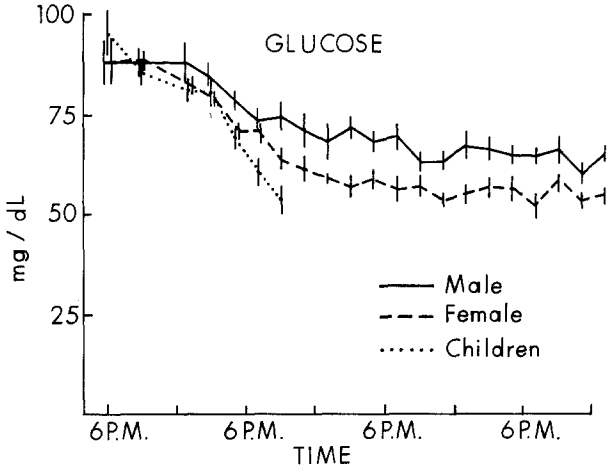

Fig. 1. Plasma glucose concentrations from men and women during 84 hr of fasting and prepubertal children during a 30 hr fast. Zero time of the fast was 1800 hr, 1 hr following their last meal. All data are expressed as the mean ± SE.

during fasting in all three groups (Fig. 2). Beta-hydroxybutyrate and acetoacetate concentrations rose more rapidly and to higher values in the children (3.75 ± 0.35 mM and 1.26 ± 0.12 mM, respectively) than in the adults (0.91 ± 0.15 mM and 0.38 ± 0.06 mM, respectively in the men and 1.68 ± 0.20 mM and 0.62 ± 0.06 mM in the women) over the initial 30 hr fasting ($p < 0.001$). The level of ketonemia observed in the children was not reached in the women until 56 hr of fasting and never achieved in the men. During the entire period of study, beta-hydroxybutyrate concentrations were higher in women than in men ($p < 0.001$) while no significant differences in acetoacetate concentrations were observed. An inverse relationship ($p < 0.001$) was observed between plasma glucose and beta-hydroxybutyrate concentrations in each group (r = 0.934, 0.925, 0.961, men, women and children, respectively).

Lactate and Pyruvate (Fig. 3)

Blood lactate concentrations were highest postprandially and decreased to similar values after 6 hr of fasting in all three groups (Fig. 3). Between 6 and 30 hr of fasting, blood lactate concentrations were higher in children than in adults ($p < 0.002$). In both men and women, lactate concentrations were similar and demonstrated a diurnal pattern with peak and nadir concentration at 1800 and 0400 hr, respectively. Because of the shorter fasting period in children, conclusions regarding a diurnal pattern could not be made.

In children, blood pyruvate concentrations were highest in the fed state and remained higher than

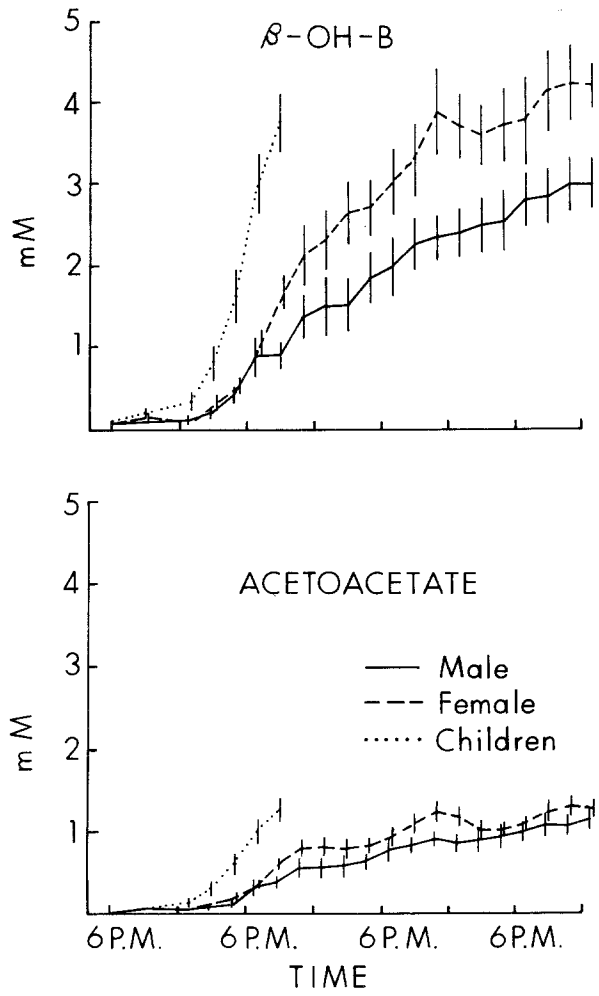

Fig. 2. Blood beta-hydroxybutyrate and acetoacetate concentrations from men and women during 84 hr of fasting from prepubertal children during a 30 hr fast. Zero time of the fast was 1800 hr, 1 hr following their last meal. All data were expressed as mean ± SE.

those observed in adults during the entire study period ($p < 0.001$). Blood Pyruvate concentrations were similar in men and women and demonstrated a diurnal variation which paralleled the fluctuation in blood lactate.

AMINO ACIDS

Alanine

Postprandial alanine concentrations (Fig. 4) were similar in the men, women and children (401 ± 17, 403 ± 33, and 447 ± 40 μM, respectively) and decreased over the initial 30 hr of fasting. The decrease in alanine concentration was greatest in the children (167 ± 17μM at 30 hr, $p < 0.01$ from the adult groups). By 30 hr fasting alanine concentrations were lower in women than men (226 ± 19 μM, women; 279 ± 16 μM, men, $p < 0.05$). Following an initial decrease in alanine over the first 30 hr of fasting in both men and women, a diurnal pattern of alanine concentrations was evident with a peak concentration at 1800 hr and nadir at 0400 hr. The patterns were identical in both men and women although the concentrations were higher in men ($p < 0.001$).

Glutamine

Postprandial glutamine concentrations (Fig. 4) were higher in men (641 ± 20 μ M) than in women and children (492 ± 20 μM and 516 ± 38 μM, respectively) and remained so throughout the period of study ($p < 0.001$). The subsequent plasma glutamine concentrations in the adults remained relatively constant; whereas a significant decrease was observed over the 30 hr of fasting in the children (516 ± 38 μ M versus 373 ± 24 μM, $p < 0.001$).

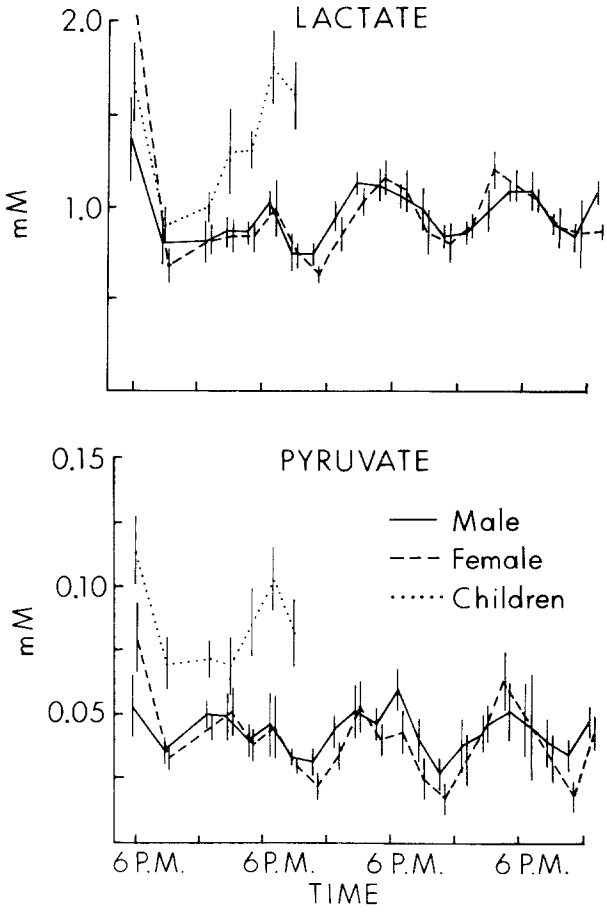

Fig. 3. Blood lactate and pyruvate concentrations from men and women during 84 hr of fasting and from prepubertal children during a 30 hr fast. Zero time of the fast was 1800 hr, 1 hr following their last meal. All data expressed as mean ± SE.

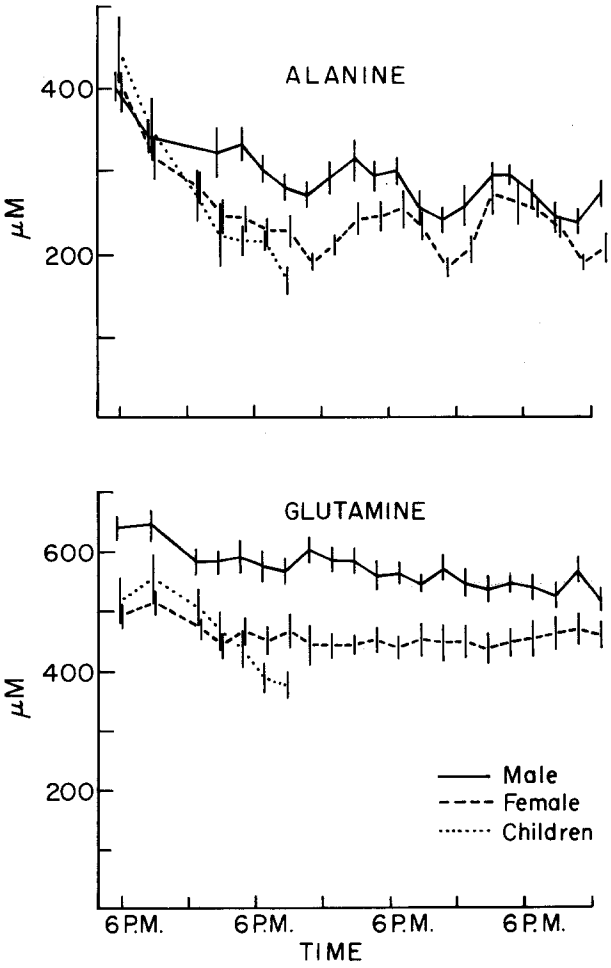

Fig. 4. Plasma alanine and glutamine concentrations for men and women during 84 hr of fasting and from prepubertal children during a 30 hr fast. Zero time of the fast was 1800 hr, 1 hr following their last meal. All data are expressed as mean ± SE.

Other Amino Acids

Total acid neutral amino acids were determined in five normal children, six men and six women. During the first 30 hr of fasting (Table 2), total amino acids (the sum of all individual amino acid concentrations) and total branched chain amino acids were determined at each sampling time. Potential gluconeogenic amino acids represent the differences between total and branched chain amino acids. At the beginning of the fast, individual amino acid concentrations (with the exception of glutamine, mentioned above) as well as the total, branched chain and potential gluconeogenic amino acid concentrations were similar in the men and women. However, the total, potential gluconeogenic and branched chain amino acid concentrations (Table 2) of the children were increased ($p < 0.05$) above adult concentrations in the postprandial period as the result of elevations of selected individual amino acids (threonine, serine, proline, alanine, valine, methionine, isoleucine and leucine). Less striking differences were observed following 14 and 22 hr of fasting. By 30 hr fasting, no significant differences in plasma amino acid concentrations were apparent between the men and women other than those of alanine and glutamine. In contrast, plasma concentrations of threonine, serine, glycine, phenylalanine and total potential gluconeogenic amino acids were lower in the children when compared to those observed in either men or women. Branched chain amino acid concentrations decreased in the children over the first 22 hr but did not change in the adults and increased in all three groups between 22 and 30 hr fasting (Table 2).

Other than plasma alanine and glutamine mentioned above, only minor differences in individual amino acids were noted between men and women during the 86 hr of fasting (Table 3). At 38 hr fasting, but at no other time during the study, plasma concentrations of serine, proline, and methionine were lower in women than in men ($p < 0.01$). From 14 hr through 38 hr fasting, both summed total amino acids and "potential gluconeogenic" amino acid concentrations in the women were lower than those observed in men ($p < 0.05$). Branched chain amino acid concentrations increased during the fast ($p < 0.01$, paired Student's t test) in both adult groups of subjects.

Hormones

Plasma insulin concentrations were similar in men, women, and children immediately following their last meal (Fig. 5). These concentrations progressively decreased over the fist 24 hr of fasting in all subjects and subsequently remained between 3 and 6 μ U/ml. Plasma insulin concentrations may have been lower in the children when compared to those of the adults over the last 8 hr of study (hr 22 through 30 of fasting) since a number of values were below the sensitivity of the insulin assay (less than 2 μU/ml).

Immediate postprandial plasma glucagon concentrations were 325 ± 36 pg/ml in women, 268 ± 30 pg/ml in children and 186 ± 21 pg/ml in men (Fig. 5). Plasma glucagon concentrations decreased to similar values by 14 hours of fasting in all three groups. Mean 0800 glucagon concentrations increased between the 38 and 86 hr of fasting (women, 131 ± 11 to 172 ± 14 pg/ml, and men, 112 ± 10 to 168 ± 10 pg/ml, $p < 0.01$ for both, paired Student's t test). No significant differences in glucagon concentrations were observed between the men and women during the 84 hr of fasting (by Student's t test or AOV).

Mean plasma cortisol concentrations were higher in women than in men during the fasting period ($p < 0.01$) (Fig. 6). Cortisol values demonstrated a signifi-

Table 2. Plasma Amino Acid Concentration in Men, Women and Children During a 30-hr Fast

Hours Fasted	0			14			22			30		
Group*	Men	Women	Children	Men	Women	Children	Men	Women	Children	Men	Women	Children
Aspartate†	12 ± 4	10 ± 2	17 ± 4	22 ± 11	12 ± 2	14 ± 2	15 ± 2	14 ± 2	11 ± 4	13 ± 2	13 ± 3	10 ± 2
Threonine	131 ± 9	119 ± 14	175 ± 12‡§	139 ± 8	115 ± 14	126 ± 12	125 ± 5	102 ± 10	81 ± 8‖	109 ± 5	97 ± 9	53 ± 5**‖
Serine	106 ± 20	138 ± 18	182 ± 13‡	142 ± 15	128 ± 8	147 ± 14	175 ± 34	128 ± 6	115 ± 11	180 ± 28	131 ± 8	78 ± 10‡**
Proline	215 ± 26	192 ± 26	362 ± 38‡§	184 ± 17	131 ± 10‡	182 ± 11§	153 ± 11	131 ± 14	130 ± 16	139 ± 18	119 ± 15	118 ± 7
Glycine	233 ± 21	186 ± 19	220 ± 27	296 ± 53	201 ± 23	194 ± 15	231 ± 12	189 ± 23	161 ± 9‡	198 ± 20	181 ± 18	92 ± 7**‖
Alanine	340 ± 38	353 ± 29	530 ± 36**‖	292 ± 24	243 ± 24	305 ± 38	281 ± 22	214 ± 17	219 ± 32	249 ± 20	200 ± 12	157 ± 14‡
Valine	258 ± 16	236 ± 19	341 ± 30‡§	258 ± 17	223 ± 15	246 ± 24	258 ± 14	211 ± 12‡	203 ± 18‡	303 ± 19	273 ± 14	291 ± 22
Methionine	34 ± 8	20 ± 5	43 ± 4	31 ± 6	19 ± 3	23 ± 3	27 ± 8	16 ± 4	16 ± 3	11 ± 2	14 ± 2	14 ± 1
Isoleucine	97 ± 13	78 ± 9	118 ± 9§	96 ± 8	71 ± 3	72 ± 5	86 ± 10	91 ± 34	71 ± 7	112 ± 8	136 ± 33	128 ± 12
Leucine	143 ± 17	126 ± 12	177 ± 10§	157 ± 18	115 ± 4	109 ± 12	135 ± 28	117 ± 10	111 ± 11	200 ± 23	167 ± 13	189 ± 17
Tyrosine	74 ± 6	72 ± 11	124 ± 23	67 ± 7	54 ± 8	58 ± 10	64 ± 4	49 ± 7	54 ± 4	63 ± 7	52 ± 5	41 ± 4‡
Phenylalanine	68 ± 6	83 ± 13	76 ± 8	67 ± 5	66 ± 12	48 ± 14‡	73 ± 7	62 ± 7	51 ± 4	73 ± 5	67 ± 8	42 ± 2§‖
Glutamine	657 ± 32	503 ± 24‖	608 ± 80	610 ± 31	466 ± 29	571 ± 61	591 ± 36	485 ± 31	526 ± 60	571 ± 30	488 ± 37	421 ± 37‡
Glutamate	44 ± 5	42 ± 4	91 ± 20	38 ± 4	47 ± 7	62 ± 16	35 ± 3	44 ± 7	55 ± 11	44 ± 7	38 ± 7	42 ± 8
Total a.a.	2315 ± 88	2155 ± 122	3181 ± 127‖**	2346 ± 116	1889 ± 72‡	2180 ± 144	2204 ± 74	1868 ± 89	1819 ± 129	2072 ± 129	1990 ± 104	1697 ± 65‡
Branched Chain	497 ± 43	440 ± 36	636 ± 48‡	510 ± 40	409 ± 23	427 ± 41	497 ± 34	419 ± 36‡	384 ± 35‡	615 ± 48	566 ± 46	608 ± 48
GNG a.a.††	1818 ± 53	1714 ± 109	2537 ± 97‖**	1835 ± 93	1480 ± 68‡	1753 ± 106	1707 ± 75	1450 ± 68‡	1435 ± 108	1457 ± 97	1424 ± 80	1089 ± 26‡‖

*Men = 6 men, women = 6 women, Children = 5 children.

†mean ± S.E., μM.

‡$p < 0.05$ from men.

§$p < 0.05$ from women.

‖$p < 0.01$ from men.

**$p < 0.01$ from women.

††Gluconeogenic amino acids = total amino acids − branched chain amino acids.

Table 3. Plasma Amino Acid Concentrations on Men and Women During an 86 hr Fast

Hours Fasted	14		26		38		50		62		74		86	
Group*	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women
Aspartate†	22 ± 11	12 ± 2	17 ± 3	11 ± 4	15 ± 3	15 ± 4	18 ± 3	20 ± 8	10 ± 1	17 ± 2‡	18 ± 3	18 ± 3	18 ± 2	18 ± 2
Threomine	139 ± 8	115 ± 14	113 ± 6	96 ± 11	113 ± 8	87 ± 10	102 ± 4	91 ± 8	109 ± 5	108 ± 7	98 ± 5	112 ± 12	100 ± 7	113 ± 10
Serine	142 ± 15	128 ± 8	166 ± 31	129 ± 6	163 ± 6	112 ± 11§	194 ± 31	120 ± 7	135 ± 10	129 ± 8	139 ± 12	142 ± 6	136 ± 9	134 ± 15
Proline	184 ± 17	131 ± 10‡	142 ± 9	119 ± 11	164 ± 6	116 ± 8§	167 ± 9	120 ± 8	170 ± 15	138 ± 7	138 ± 12	155 ± 15	180 ± 8	147 ± 24
Glycine	296 ± 53	201 ± 23	211 ± 13	178 ± 18	198 ± 12	167 ± 24	196 ± 9	160 ± 9	168 ± 14	168 ± 12	184 ± 13	202 ± 14	182 ± 13	195 ± 29
Alanine	292 ± 24	243 ± 24	257 ± 20	210 ± 17	254 ± 24	175 ± 21‡	271 ± 16	228 ± 30	230 ± 28	186 ± 22	245 ± 16	239 ± 20	246 ± 17	188 ± 28
Valine	258 ± 17	223 ± 15	281 ± 20	229 ± 10	404 ± 30	356 ± 23	391 ± 31	373 ± 35	501 ±l 32	431 ± 38	425 ± 46	428 ± 29	462 ± 37	414 ± 54
Methionine	31 ± 6	19 ± 3	25 ± 5	12 ± 3	21 ± 2	11 ± 2§	21 ± 1	20 ± 5	20 ± 9	23 ± 6	17 ± 3	23 ± 4	32 ± 9	25 ± 8
Isoleucine	96 ± 8	71 ± 8	91 ± 5	107 ± 32	146 ± 12	148 ± 23	143 ± 10	159 ± 36	173 ± 8	177 ± 16	151 ± 14	154 ± 15	179 ± 18	153 ± 23
Leucine	158 ± 18	115 ± 4	172 ± 18	134 ± 9	279 ± 23	239 ± 20	255 ± 20	226 ± 18	322 ± 21	308 ± 21	270 ± 25	265 ± 20	315 ± 18	276 ± 39
Tyrosine	67 ± 7	54 ± 8	64 ± 5	129 ± 13‡	67 ± 6	53 ± 7	73 ± 5	58 ± 7	74 ± 4	64 ± 8	67 ± 4	62 ± 6	81 ± 2	65 ± 10
Phenylalanine	67 ± 5	66 ± 12	69 ± 5	61 ± 8	71 ± 7	61 ± 6	78 ± 7	67 ± 7	69 ± 3	66 ± 5	69 ± 4	67 ± 5	71 ± 1	64 ± 10
Glutamine	610 ± 31	466 ± 29‡	587 ± 41	485 ± 31	598 ± 25	438 ± 30§	567 ± 19	458 ± 30‡	561 ± 34	476 ± 38	528 ± 26	459 ± 29	530 ± 21	480 ± 28
Glutamate	38 ± 4	47 ± 7	35 ± 4	39 ± 7	32 ± 1	31 ± 4	31 ± 4	33 ± 3	31 ± 4	31 ± 3	31 ± 4	30 ± 3	29 ± 2	28 ± 3
Total a.a.	2346 ± 116	1889 ± 72‡	2201 ± 71	1873 ± 78‡	2464 ± 116	2028 ± 106‡	2434 ± 77	2151 ± 133	2525 ± 41	2321 ± 131	2435 ± 139	2352 ± 91	2554 ± 114	2238 ± 225
Branched China a.a.	510 ± 40	409 ± 23	544 ± 42	470 ± 32	829 ± 63	743 ± 56	789 ± 58	758 ± 83	995 ± 5	915 ± 69	847 ± 84	847 ± 63	956 ± 70	940 ± 57
"GNG"-a.a.‖	1835 ± 93	1480 ± 68‡	1657 ± 84	1404 ± 59‡	1635 ± 104	1285 ± 87‡	1645 ± 86	1393 ± 65	1530 ± 109	1406 ± 70	1589 ± 68	1505 ± 34	1598 ± 55	1299 ± 208

*Men, $n = 6$; women, $n = 6$.
†Mean ± SE, μM.
‡$p < 0.05$ from men.
§$p < 0.01$ from men.
‖Gluconeogenic amino acids = total amino acids − branched chain amino acids.

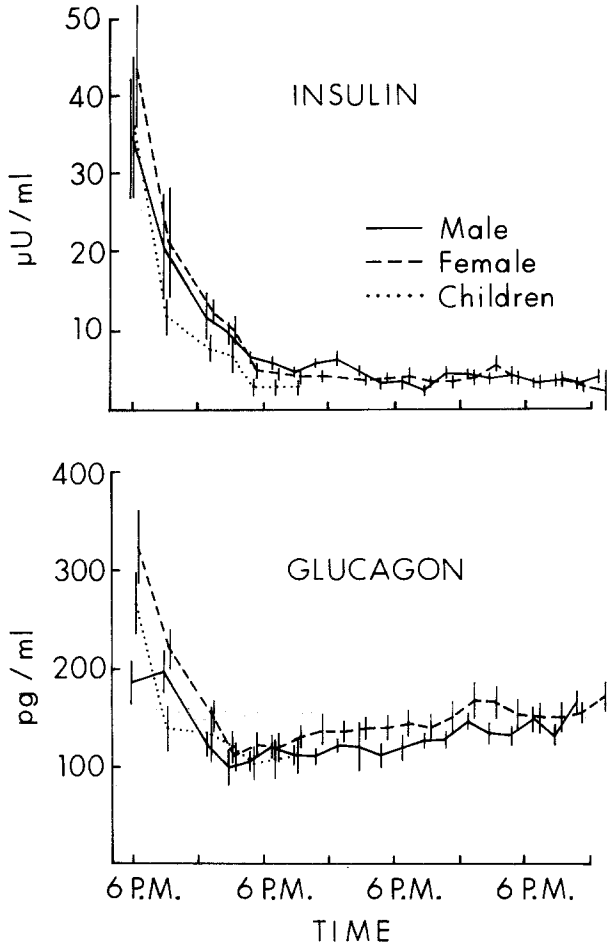

Fig. 5. Plasma insulin and glucagon concentration from men and women during 84 hr of fasting and from prepubertal children during a 30 hr fast. Zero time of the fast was 1800 hr, 1 hr following their last meal. All data are expressed as mean ± SE.

cant diurnal variation with fasting in men and women with peak concentrations at 0800 hr and with the lowest concentrations observed between 2100 hr and midnight. In children, a peak concentration in cortisol was observed after the initial overnight fast with a second peak occuring between 2000 and 2400 hr (after 26–30 hr of fasting).

Plasma growth hormone concentrations were similar in all three groups during the first 30 hr of fasting (Fig. 6). During the last 54 hr of fasting a clear diurnal pattern was observed in both men and women. Peak plasma growth hormone concentrations occurred at 2400 hr (14–17 ng/ml in the females and 10–15 ng/ml in the males) with the lowest concentrations observed at 0800 hr (3–5 ng/ml in females and 1–3 ng/ml in males).

DISCUSSION

Adaptation to starvation is dependent upon a complex of inter-related mechanisms involving hormonal regulation of substrate mobilization, interconversion and utilization. The present study demonstrates that the general pattern in circulating substrates and hormones in response to fasting are similar among fasted men, women and children, but the relative rates of change and absolute concentrations achieved are different.

Plasma insulin concentrations were similar among the men, women and children during their respective periods of study. Since many of the insulin concentrations obtained from the children at the later time points in their fasts were below the sensitivity of the assay (less than 2 μU/ml), values reported in the children may have in fact been lower than those of adult subjects. Although mean glucagon, growth hormone and cortisol concentrations were higher in women, only the cortisol concentrations were significantly different ($p < 0.001$). Since four of the women had been taking oral contraceptives prior to the study,

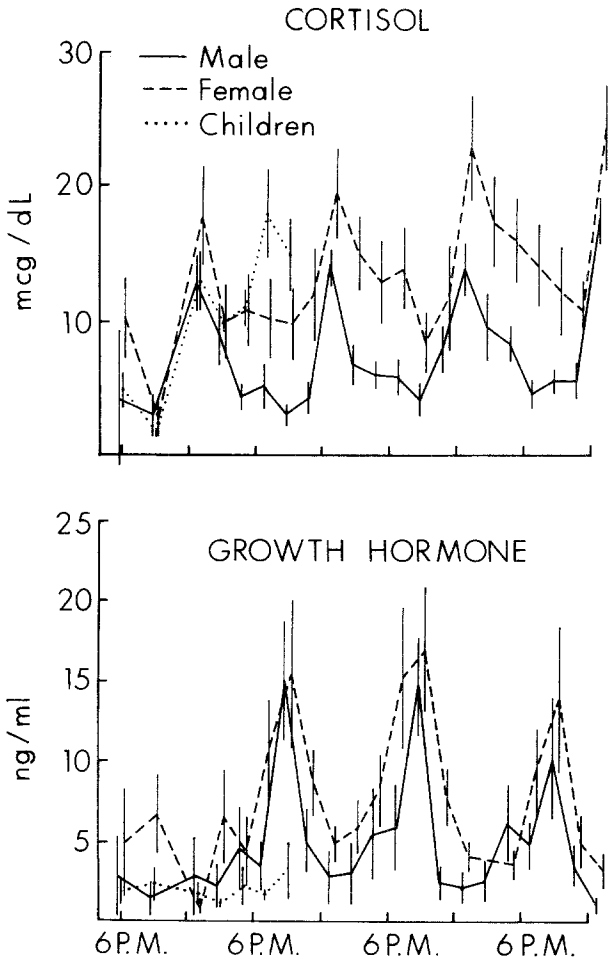

Fig. 6. Plasma cortisol and growth hormone concentrations from men and women during 84 hr of fasting and from prepubertal children during a 30 hr fast. Zero time of the fast was 1800 hr, 1 hr following their last meal. All data are expressed as mean ± SE.

this difference may be secondary to estrogen induced increases in cortisol binding protein concentrations; however, no differences were observed in peak or nadir cortisol values between those taking oral contraceptives and those who were not. The disruption of the normal circadian rhythm of cortisol in the children may reflect greater metabolic "stress" in these subjects when compared to adults. The mean plasma glucagon concentrations rose with fasting, but were not significantly different between men and women; the absence of difference between plasma glucagon values in men and women is in agreement with recent studies[6] and is to be contrasted to earlier reports.[4,5]

The greater degree of ketonemia in women when compared to men is consistent with a higher free fatty acid concentration previously observed in fasted women.[4] The striking ketosis which occurred in the children may be related to subtle differences in insulin concentrations during the fast, or increased plasma cortisol[17] and/or growth hormone[18] resulting in both increased mobilization of free fatty acids and increased ketone body production. The greater degree of ketosis observed in the women when compared to men cannot be attributed to differences in circulating insulin, growth hormone or glucagon. However, since administration of glucocorticoids to normal and diabetic volunteers results in increased ketone body concentrations;[17,19] the relative hyperketonemia observed in women may be the result of increased circulating concentrations of cortisol. However, the absence of any clear diurnal variation in blood beta-hydroxybutyrate or acetoacetate would not support a predominant role of cortisol in these differences.

In contrast to the relative small differences in glucoregulatory hormone concentrations, remarkable differences in circulating concentrations of potential gluconeogenic amino acids were observed among the three groups studied. Alanine and glutamine are two important gluconeogenic substrate preferentially released by muscle.[20–22] Availability of these gluconeogenic substrates has been associated with abnormalities of glucose homeostasis in both children and adults.[2,3,7–9]

In the present study, both the rate of fall and the absolute decrease in alanine concentrations during the first 30 hr of fasting are greater in children than in adults; over the entire study, plasma alanine values were lower in women than men. The decrease in plasma alanine concentrations during an infusion of beta-hydroxybutyrate, has lead to speculation that ketone body concentrations may decrease alanine efflux from muscle[23] and could explain the relative differences in alanine concentration among the groups studied. However, other evidence does not support this hypothesis. During the first 30 hr of fasting in the present study, the plasma alanine concentrations were similar between the women and children, despite significantly greater degree of ketosis in the children; and during the last 24 hr of the fast, plasma alanine concentrations were not significantly different between men and women despite a significantly higher concentration of beta-hydroxybutyrate in the female subjects. In addition, infusion of betahydroxybutyrate in normal adults results in an increase in alanine production,[24] and an even greater increase in alanine utilization resulting in a decrease in its plasma concentration. Therefore, ketosis, per se, may not play a predominant role in the regulation of circulating alanine concentrations.

The differences in circulating concentrations of alanine between the men and women are to be contrasted with the nearly identical plasma lactate and pyruvate concentrations observed. In contrast, the initial venous lactate and pyruvate concentrations were highest in the children and increased even further over the last 6 hr of fasting. Assuming normal hepatic uptake of these substrates, these data suggest that lactate production may be accelerated in pediatric aged patients during caloric deprivation.

The relationship between glucocorticoids and plasma alanine, lactate and pyruvate concentrations deserves comment. Since pyruvate is a common precursor of both alanine and lactate, it is of interest to note that the circadian rhythm of plasma alanine, but not glutamine, paralleled those of lactate and pyruvate, and the nadir concentrations of these substrates was 0400 hr, the nadir of plasma cortisol. Administration of pharmacologic[7] and physiologic[3] doses of glucocorticoids has been associated with increased plasma alanine and improved fasting glucose homeostasis in children with ketotic hypoglycemia or panhypopituitarism. In addition, infusion of cortisol at a rate sufficient to increase plasma cortisol within the physiologic range increases alanine production within 120–150 min.[25] Therefore, circulating cortisol may play a physiologic role in the modulation of production or utilization of these important potential gluconeogenic substrates.

The primary source of circulating glutamine is skeletal muscle,[21,22,26] and the two major sites of its catabolism are the kidney (gluconeogenesis and ammoniagenesis)[27,28] and gut (with resultant production of alanine)[29,30] therefore, the differences in plasma glutamine among the groups studied must be a result of differences in either production or utilization. Increased rates of utilization of glutamine might

contribute to the lower plasma concentrations observed during the evolution of fasting ketosis (and presumed mild metabolic acidosis) in children, since acidemia increases renal extraction of glutamine, providing substrate for both gluconeogenesis and ammoniogenesis.[27,28] Such an argument would not explain the differences in glutamine concentrations observed between men and women or children during the early hours of fasting. Since men have a greater muscle mass than the women or children, both the increased plasma alanine and glutamine could be the result of increased release of these potential gluconeogenic amino acids. Increased availability of potential gluconeogenic substrates during fasting may be partially responsible for the higher plasma glucose concentrations observed in men when compared to women or children.

The present studies demonstrate clear differences in circulating concentrations of potential gluconeogenesis substrates among men, women and children. The increase rate of glucose utilization when expressed as mg/kg · min previously reported in children[31] may result in the lower plasma glucose and insulin concentrations and increased cortisol concentrations. Under these metabolic circumstances, increased Cori cycle activity might be favored, accounting for the increased venous lactate and pyruvate concentrations. Whether the differences in circulating concentrations of alanine and glutamine between men and women (and children) are the result of differences in body composition, glucose requirements or other metabolic or hormonal factors cannot be determined by the present study; however, these data suggest that differences in the metabolic adaptation to caloric deprivation between men and women may be secondary to relative availability of potential gluconeogenic substrates.

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